SCALING ANALYSIS AND DEVIATION EVALUATION OF NATURAL CIRCULATION IN SIMPLIFIED LEAD-BISMUTH FAST REACTOR

Jianing Xu[1,2], Zhen Wang[1 (Corresponding author)], Rui Pan[1], Shichao Zhang[1], Qiusun Zeng[1], Jie Yu[1]
[1]Hefei Institutes of Physical Science, Chinese Academy of Sciences, Hefei, 230031, China
[2]University of Science and Technology of China, Hefei 230026, China

Email contact of corresponding author: zhen.wang@inest.cas.cn

Abstract

Lead-Bismuth Eutectic (LBE) is an attractive coolant for the Generation-IV advanced reactor systems. Its outstanding natural circulation capability enhances the system safety of the reactors. Lead-bismuth fast reactors have broad application prospects in areas such as miniaturization and specialized uses. The scaled experimental facility will help analyze the flow characteristics of the liquid LBE. The scaling theory is used to evaluate whether the experimental device can reproduce the key characteristics in the prototype reactor. To study the natural circulation phenomenon in the lead-bismuth fast reactors, a scaling method for a simple lead-bismuth natural circulation system was derived based on the Hierarchical Two-Tiered Scaling (H2TS) theory. The natural circulation model was calculated using the CFD software Fluent. The study focuses on the relationship between the flow area of the circulation structure and the transient scaling distortion. The reliability of the scaling method in the field of natural circulation scaling for lead-bismuth reactors was evaluated. Results show that the scaling method exhibits rationality in the steady circulation of LBE. Meanwhile, the characteristic parameters of the scaled-down models are in good agreement with the theoretical values. For the transient processes, the LBE circulation flow area influences the dynamic deviation, but the overall deviation is within a reasonable range. This research has practical significance for the design of scaled experimental facilities for lead-bismuth reactors.

1. INTRODUCTION

Liquid lead-bismuth eutectic (LBE) is one of the ideal coolants for Generation-IV fast reactors[1]. The safety of the reactors under the accident conditions can be enhanced by its excellent natural circulation capacity. Natural circulation can also drive the coolant flow in some reactors under the normal operation[2,3]. Therefore, the research on the natural circulation characteristics of the LBE is crucial to the development of the fast reactors. It is costly and impractical to conduct thermal-hydraulic experiments using the full-scale lead-bismuth reactors. Scaling analysis theory provides a theoretical basis for the design of scaled-down facilities. It is of great significance for conducting economical and feasible experimental studies on the LBE flow characteristics.

In the past, scaling analysis on the traditional water reactors has been conducted in depth, and a wealth of scaling analysis methods have been developed[4,5]. Lu et al.[6] derived the scaling criteria applicable to both single- and two-phase water circulation systems. Li et al.[7,8] discussed the draining process of the core makeup tank and proposed the corresponding scaling method to evaluate the dynamic characteristics. Cheng et al.[9] conducted an experimental study based on a reduced-height water facility to provide the data support for the improvement of the scaling method. In contrast, research on scaling analysis methods for the phenomena of LBE fluid is relatively lacking. In recent years, the scaling method has been gradually applied to the development of advanced reactors. Su et al.[10] derived a scaling method for the thermal stratification of the LBE in the Accelerator Driven System (ADS) reactor based on the H2TS theory. The geometric model of the stratified space was obtained. Zhao et al.[11,12] performed the scaling analysis on the Small Natural circulation Lead-based Fast Reactor (SNCLFR-10), and discussed the steady scaling deviation using a two-dimensional LBE loop model.

In this study, the scaling analysis criteria for the lead-bismuth natural circulation were derived by applying the H2TS theory. Furthermore, considering the difficulty in conducting pool-type experiments, a group of simplified criteria was obtained for comparison to facilitate the design of the experimental loop. Various prototype loops were set by varying the diameter parameters. The correlation between the flow area of the circulation system and the scaling distortion was discussed. The Fluent code was used to calculate the transient cases of the numerical models, and the rationality of the proposed method was verified.

2. SCALING ANALYSIS METHOD

The schematic diagram of the coolant flow in the lead-bismuth fast reactors is shown in Fig. 1. The highlighted section depicts a typical natural circulation process. When the pump shuts down, the liquid LBE absorbs the heat in the core and then

flows upward into the upper plenum. Subsequently, the heat is transferred to the secondary side in the heat exchanger. The fluid enters the lower plenum and finally returns to the core. The marked circulation in Fig.1 was simplified as a single natural circulation loop for the subsequent analysis.

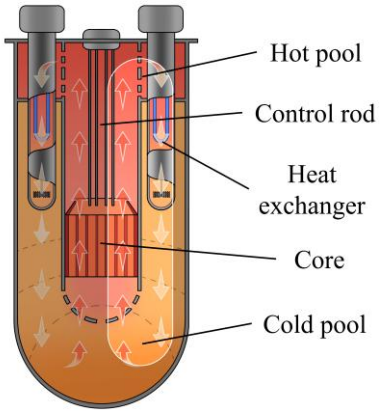


FIG. 1. Schematic diagram of natural circulation process in LBE fast reactors.

The continuity equation, momentum equation, and energy equation of the above loop can be expressed as:

$$\rho_i u_i A_i = \rho_0 u_0 A_0 \tag{1}$$

$$\frac{d}{dt}\left[\rho u \sum_{i=1}^{N}\left(l\frac{A_0}{A}\right)_i\right] = \beta\rho g H \Delta T - \frac{1}{2}\rho u^2 \sum_{i=1}^{N}\left[\left(\frac{A_0}{A}\right)^2\left(\frac{fl}{d}+k\right)\right]_i \tag{2}$$

$$\rho C_p\left(\frac{\partial T}{\partial t}+u\frac{\partial T}{\partial z}\right)=\frac{4q}{d} \tag{3}$$

$$\rho C_p\left(\frac{\partial T}{\partial t}+u\frac{\partial T}{\partial z}\right)=\frac{4h}{d}\left(T_s-T\right) \tag{4}$$

$$h\left(T_s-T\right)=-k_s\frac{\partial T_s}{\partial y} \tag{5}$$

Here, ρ is the density of the LBE fluid, u is the flow rate, and A represents the flow area of the loop. Subscripts i and 0 denote the parameters at any arbitrary and the reference position in the loop, respectively. l and H are the length and the height of the circulation, β is the thermal expansion coefficient of the LBE, T and T_s represent the coolant and wall temperature, k is the form loss coefficient, f is the friction loss coefficient, k_s is the thermal conductivity of the solid wall, q is the heat flux of the core, and h is the convection heat transfer coefficient.

Thus, the following criteria can be obtained by solving the above equations simultaneously.

$$\prod_{Ri,R}=\left(\frac{\beta g H\Delta T}{u^2}\right)_R=\left(\frac{l\Delta T}{u^2}\right)_R=1 \tag{6}$$

$$\prod_{Fri,R}=\left(\frac{fl}{d}+k\right)_R=1 \tag{7}$$

$$\prod_{Qs,R}=\left(\frac{ql}{\rho C_p du\Delta T}\right)_R=\left(\frac{ql}{du\Delta T}\right)_R=1 \tag{8}$$

$$\prod_{st,R} = \left(\frac{k_s l}{\rho C_p du\delta} \right)_R = \left(\frac{l}{du\delta} \right)_R = 1 \tag{9}$$

δ in Eq. (9) is the wall thickness of the loop. In actual conditions, wall thickness is a parameter that cannot be arbitrarily adjusted. Therefore, it is not required to strictly consist with the criteria. It is assumed that the fluid thermophysical properties of the prototype are identical to those of the model. After selecting the length scale ratio $l_R = c$, the scaling criterion array for the LBE natural circulation can be obtained. Where the friction coefficient can be substituted by using $f = 0.3164/Re^{1/4}$.

$$m_R = c^{19/10} \tag{10}$$

$$d_R = c^{7/10} \tag{11}$$

$$q_R = c^{1/5} \tag{12}$$

For experimental studies of the LBE natural circulation, the loop-type facilities are commonly used. The d_R of the experimental loop cannot always strictly follow the above criteria. Instead, it is determined based on the practical conditions. Given that $d_R = \xi$, a set of simplified and flexible criteria can be obtained.

$$m_R = c^{1/2}\xi^2 \tag{13}$$

$$q_R = c^{-1/2}\xi \tag{14}$$

Therefore, the lead-bismuth circulation with a large flow area, derived from the pool-type reactor, will be scaled to a simplified experimental loop. Obviously, the heat flux ratio of the scaled-down model decreases as the ratios of diameter and flow area decrease. The relatively small core power is required in the scaled-down experimental loops, thereby greatly facilitating the facility design. This may lead to additional scale distortion, which needs to be fully considered in practical applications. A comparative analysis of the two criteria will be presented in the subsequent sections.

3. MODEL AND VALIDATION

3.1. Numerical model and boundary conditions

A set of numerical models for the lead-bismuth natural circulation loop, as shown in Fig. 2, was established and calculated using the Fluent code. It is mainly divided into two parts, the prototype and the scaled-down models. Depending on the differences in scaling criteria, the scaled model is further classified into the standard model and the simplified mo del. The different diameters are selected to analyze the influence of the flow area on the scaling process. The prototype loops were designed with reference to the KYLIN-II lead-bismuth natural circulation experimental loop of the Institute of Nuclear Energy Safety Technology, Hefei Institutes of Physical Science, Chinese Academy of Sciences [13]. The main design parameters of the prototype loops are presented in Table 1. Furthermore, when the LBE circulation stabilized, the power was increased by 10% to create a transient process until the circulation re-stabilized.

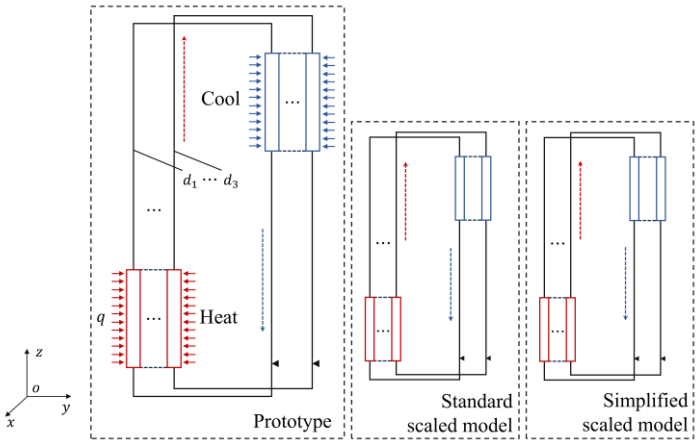


FIG. 2. Numerical model of LBE natural circulation loops.

TABLE 1. PROTOTYPE LOOPS PARAMETERS

	Prototype loop 1	Prototype loop 2	Prototype loop 3
Total height (m)		5	
Pipe inner diameter (mm)	26	35	50
Heating section length (m)		1.8	
Cooling section length (m)		1.5	
Height between cooling and heating sections (m)		3.35	
total loop length (m)		12	
Initial input power (kW)		0-30	

The typical length scale ratio $l_R = 0.5$ was selected and the criterion arrays of the standard and simplified scaled-down loops can be obtained separately, as shown in Table 2. Three standard scaled loops have a uniform scaling ratio. Correspondingly, the simplified scaled loops have the same geometric parameters. To facilitate comparison, the geometric parameters of the simplified scaled loops are consistent with those of the standard scaled loop corresponding to Prototype 1.

TABLE 2. SCALING CRITERION BASED ON THE STANDARD AND SIMPLIFIED CRITERIA

	Scaled loop number	Scaling ratio				
		l_R	d_R	m_R	ΔT_R	q_R
Standard model	Loop 1-3	0.5	0.616	0.268	1	0.871
Simplified model	Loop 1	0.5	0.616	0.268	1	0.871
	Loop 2	0.5	0.457	0.148	1	0.647
	Loop 3	0.5	0.320	0.072	1	0.453

The thermal properties of liquid lead-bismuth are provided in the following equations[14].

$$\eta = 4.94\times10^{-4}\exp\left(754.1/T\right) \tag{15}$$

$$\lambda = 3.284 + 1.617\times10^{-2}T - 2.305\times10^{-6}T^2 \tag{16}$$

$$\rho = 11096 - 1.3235T \tag{17}$$

$$C_p = 164.8 - 3.94\times10^{-2}T + 1.25\times10^{-5}T^2 - 4.56\times10^{5}T^{-2} \tag{18}$$

Where η is the viscosity and λ is the thermal conductivity of LBE. In the heating section, a heat flux boundary condition was used to simulate the core, while a constant wall temperature boundary condition was used for the cooling section to simulate the heat exchanger. In addition, other sections of the loop were assumed to have the effective thermal insulation performance, and thus the adiabatic wall boundary condition was applied. The Boussinesq hypothesis[15] was applied to the LBE fluid. Furthermore, the recent studies indicate that the predictions of the SST k-omega model are in good agreement with the experimental correlations, especially in the low flow rate region corresponding to the LBE natural circulation[16]. Therefore, the SST k-omega model was adopted in the calculation.

3.2. Numerical method verification

The geometric parameters of Prototype loop 1 are consistent with those of the KYLIN-II LBE natural circulation loop. Therefore, the reliability of the numerical method was validated through comparison with the experimental data. Firstly, the grid independence analysis was performed for the loop. As shown in Fig. 3a, when the grid number reaches 1×10^6, the changes in circulation flow rate and temperature difference tend to stabilize. To balance the computational time and accuracy, the grid number for the loop was determined to be one million.

Fig. 3b and Fig. 3c show the trends of the mass flow rate of LBE circulation and the temperature difference between the hot and cold sections with the power of the heating section, respectively. The simulation results were compared with those from

the references[13]. The curves indicate that the simulation results show good agreement with the experimental data, except when the power is less than 5 kW. When the power is relatively small, the experimental values of the circulation flow rate are lower than the computational values, while the temperature difference is higher. It is caused by the unavoidable heat dissipation of the heat pin in the actual experimental loop[13]. The results demonstrate that the modeling approach used in the study is reasonable.

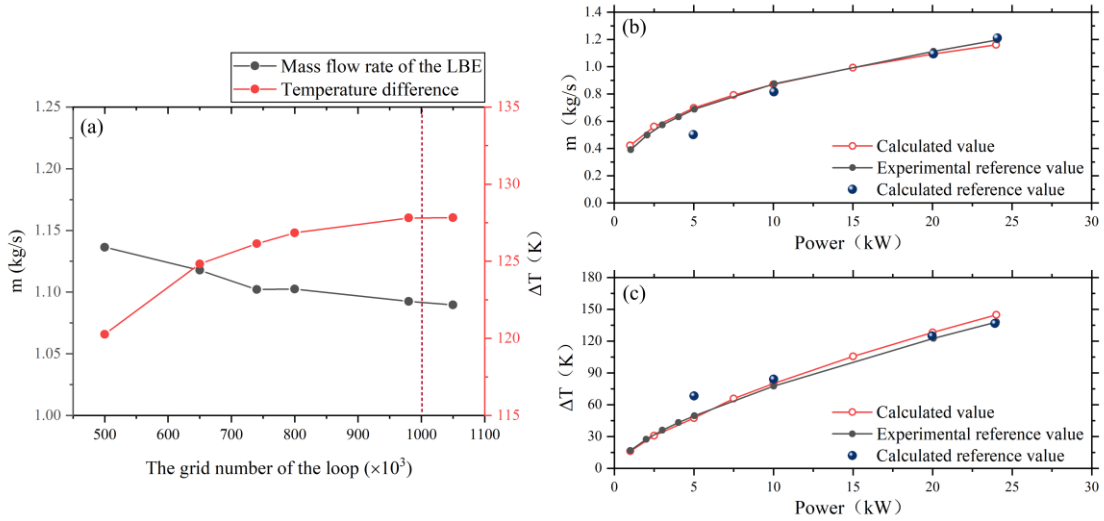


FIG. 3. a. Relationship between the flow characteristic parameters and the number of mesh. b. Power- Flow rate curve of the computation and experiment. c. Power- Temperature difference curve of the computation and experiment.

4. RESULTS AND ANALYSIS

Fig. 4 shows the steady errors of the LBE flow rate and the temperature difference between the hot and cold sections. The horizontal axis represents the diameter of the prototype loops. Different powers of the core were set for each prototype for comprehensive comparison. For the standard scaling models, the steady deviation of the mass flow rate is less than 7.3% under all cases, while the deviation of the temperature difference remains below 7.5%. The difference in the flow area of the prototype cases has no significant effect on the distortion, and all scaled-down models accurately simulate the steady characteristic of the prototype circulation. In contrast, for the simplified scaling models, the distortion increases with the flow area of the loop. The scaling of the diameter results in the distortion of the friction number of the criteria. In practice, calibration of the resistance before experiments can be performed to meet the friction criterion and avoid the excessive distortion. This can be adjusted by the electric regulating valve in the loop[9,17,18]. When the flow area of the prototype loop is fixed, the distortion associated with both methods increases as power rises, although the trend is relatively insignificant.

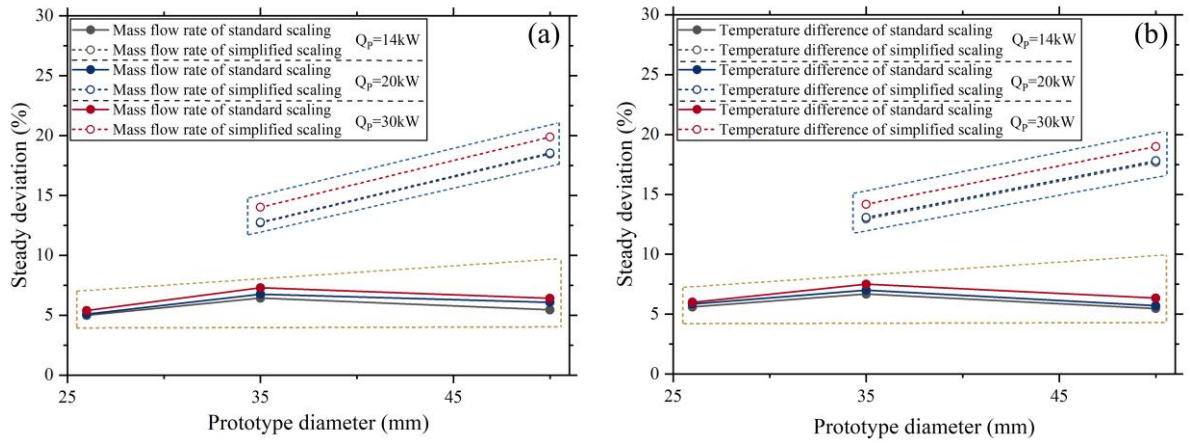


FIG. 4. a. Steady error of mass flow rate. b. Steady error of temperature difference.

Fig.5 shows the transient response characteristics of the LBE natural circulation. To facilitate comparative analysis, all parameters represented by the vertical axis were normalized. The horizontal ordinate in the figure is defined as $t^* = t/t_0$. Subscript 0 is the reference value, representing the time required for the liquid LBE to complete a full cycle in the loop. Fig.5a-c show the transient mass flow curves of the prototype and the scaled-down models based on the standard and simplified criteria.

The ordinate was defined as $(m-m_0)/(m_1-m_0)$. Subscripts 0 and 1 represent the initial reference value and the final steady parameter, respectively. Overall, when the power changes, the transient responses of the mass flow rate under all cases can be roughly divided into three stages. The circulation flow rate first rose rapidly to a maximum value over time, then entered a fluctuating phase lasting 0.5 to 3 circulation cycles. Finally, the second steady state was reached. Obviously, all cases exhibit considerable consistency during the rise of the core power. The differences between the curves are mainly reflected in the fluctuations of the transient response. For the prototype cases, the amplitude of fluctuation increases with the rise in power and the loop diameter, and the influence of diameter is more significant. Meanwhile, the duration of the fluctuation is prolonged only with the increase in diameter. Regardless of the diameter of the prototype loop, the scaled-down cases based on the standard scaling criteria can accurately simulate the transient response of the prototype, especially the fluctuation stage of the flow rate. However, the fluctuation amplitude of the curves under the scaled-down cases is smaller than that of the prototype.

Correspondingly, the fluctuation stage of the simplified scaled cases is significantly briefer than that of the prototype, especially when the diameter ratio ξ is relatively small. It can be easily explained that even though the several prototype loops have the different diameters, their simplified scaled models have the same geometric parameters. The only difference lies in the heat flux. After applying the simplified scaling method, an experimental loop can be used to simulate the prototype facility, which has a relatively large flow area. The LBE in the scaled-down loop exhibits an approximately one-dimensional flow, which means that it is impossible to accurately simulate the small-amplitude fluctuation transient responses of the prototype.

Fig.5d-f show the normalized temperature difference in the hot and cold sections of the model loop in the transient process. Similarly, the ordinates are defined as $(\Delta T-\Delta T_0)/(\Delta T_1-\Delta T_0)$. When the power changes, the change of the temperature difference is significantly faster than that of the flow rate. This is because the fluid temperature in the hot section changes immediately with the power. Overall, the response patterns of the temperature difference curves corresponding to all scaled-down cases are almost consistent with those of the flow rate curves.

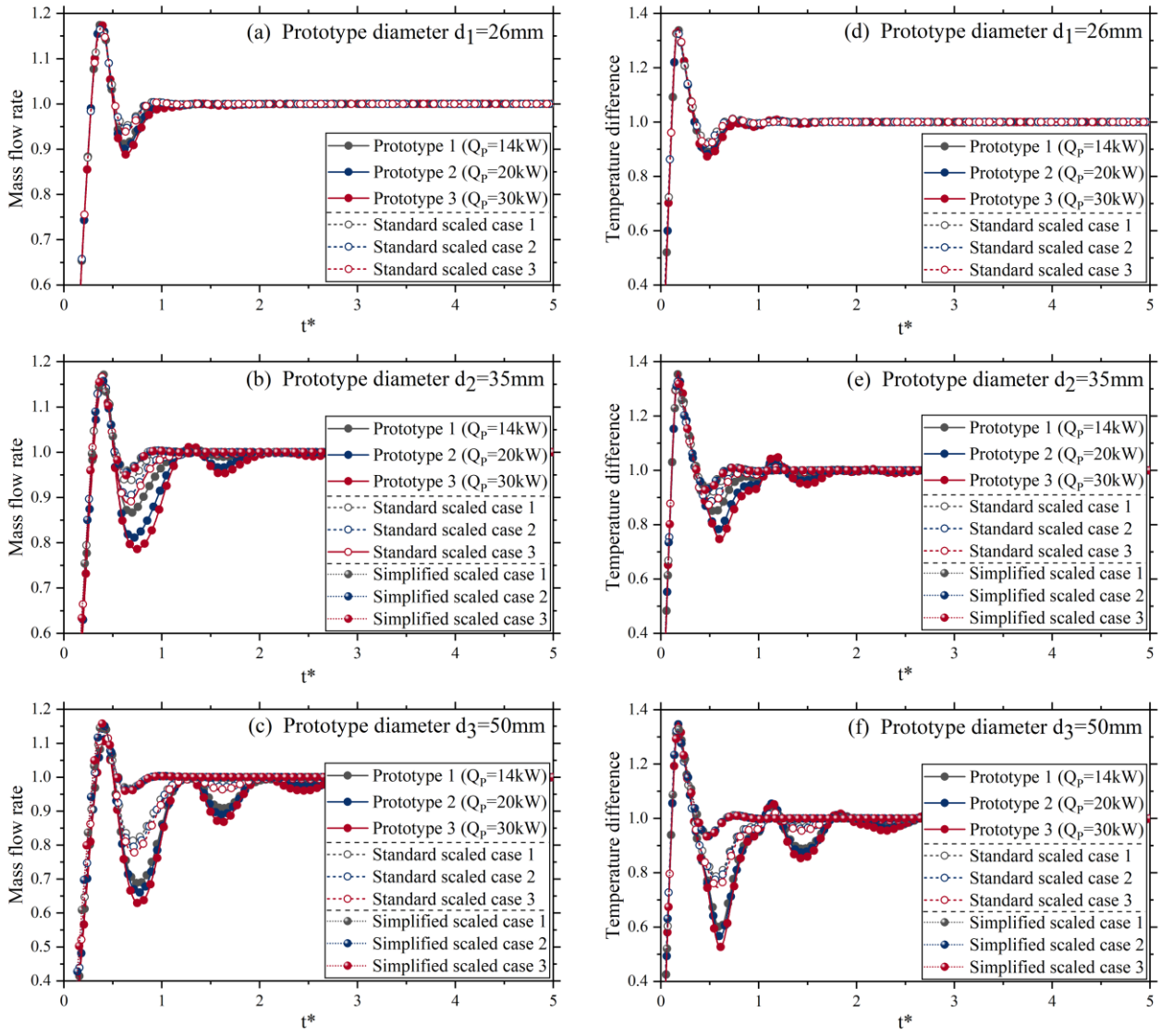


FIG. 5. a-c. normalized mass flow rate curve. d-f. steady error of temperature difference.

Fig. 6a-c show the transient relative errors in the normalized mass flow rate while Fig. 6d-f show those in the temperature difference. Except during the initial stages of the core power variation, the maximum error occurs at the first peak of the flow rate fluctuation. For a fixed prototype diameter, the error of the LBE flow rate increases slightly with the core power. For a fixed core power, the error increases as the prototype diameter increases. Under the same prototype cases, the transient response deviation of the simplified scaled-down models during the flow fluctuation is larger than that of the standard scaled-down models. Compared with the mass flow rate, the temperature difference exhibits a larger transient deviation as the diameter increases.

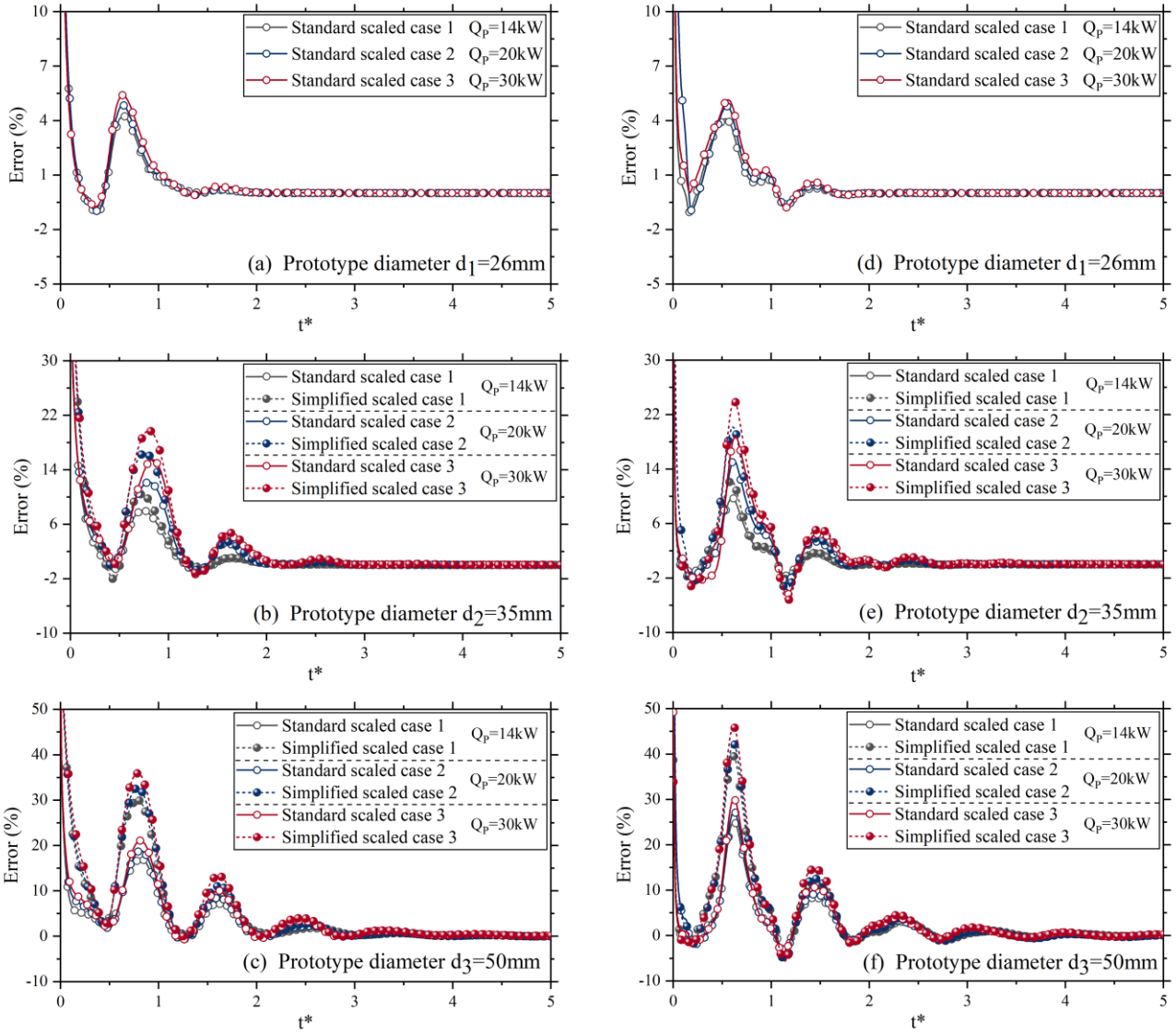


FIG. 6. a-c. transient error of mass flow rate. d-f. transient error of temperature difference.

5. CONCLUSION

This study performed a scaling analysis on the transient response characteristics of LBE natural circulation and obtained a set of scaling criteria. Furthermore, a group of simplified similarity criteria was developed to simplify the analysis. Three prototype loops with different flow areas and their corresponding scaled-down models were designed. The scaling method was verified using the Fluent code. The conclusions are as follows:

(1) When strictly adhering to the scaling criteria, the flow area ratio of the scaled-down model is determined by the length scaling ratio. The scaling distortion of steady circulation parameters is less than 7.5%, and is independent of the flow area of the LBE loop. The simplified criteria allow more flexible flow area scaling, which facilitates the experimental loop design. However, the scaling distortion increases as the area ratio decreases. The resistance coefficient calibration of the scaled-

down facility should be conducted to satisfy the similarity of criterion numbers.

(2) The standard scaling method can accurately reproduce the transient response process of the prototype lead-bismuth natural circulation, including both the initial phase and the subsequent fluctuation phase of the characteristic parameters. The simplified scaling method exhibits scaling distortion during the fluctuation phase of the circulation flow rate and temperature difference. However, the transient deviations in both parameters increase with the larger flow area and the higher power, and the flow area of the prototype loop is a more dominant factor.

REFERENCES

[1] He, J., Huang, Q.Y., Zhu, Z.Q., et al. Experiment and analysis on flow rate and temperature of liquid PbBi in PREKY. Nucl. Sci. Tech. (2016) 27(2): 48.

[2] Chen, H.L., Chen, Z., Zhou, T., et al. Preliminary thermal-hydraulic design and analysis of china lead alloy cooled research reactor (CLEAR-I). Proceedings of the 9th International Topical Meeting on Nuclear Thermal-Hydraulics, Operation and Safety (NUTHOS-9' 12). (2012) 1-9.

[3] Chen, H.L., et al. Conceptual design of a small modular natural circulation lead cooled fast reactor SNCLFR-100. International journal of hydrogen energy (2016). 41.17: 7158-7168.

[4] Xu, J.N., Li, X.B., Liu, Y.S., et al., Two-parameter dynamical scaling analysis of single-phase natural circulation in a simple rectangular loop based on dilation transformation. Nucl. Sci. Tech. (2022) 33, 156.

[5] Xu, J.N., Li, X.B., Wang, Z.Y., et al., Dynamic scaling characteristics of single-phase natural circulation based on different strain transformations. Nucl. Sci. Tech. (2023) 34(9), 142.

[6] Lu, D.H., Xiao, Z.J., Chen, B.D. Investigation on Basic Equations and Scaling Criteria of PWR Natural Circulation. Nuclear Power Engineering. (2009) 30(03): 72-84+94.

[7] Li, X.B., Zhan, B.T., Liu, Y.S., et al. Dynamic characteristics and scaling distortion of gravity-driven draining process in Core Makeup Tank. Ann. Nucl. Energy. (2021) 151(4): 107963.

[8] Li, X.B., Li, N., Wu, Q., et al., Application of dynamical system scaling method on simple gravity-driven draining process. Nucl. Sci. Technol. (2017) 55, 11-18.

[9] Cheng, C, Lu, D.H., Qianhua Su, Q.H., et al. Experimental study on flow rate distortion of natural circulation reduced-height scaling. Nucl. Eng. Des. (2023) 402: 112122.

[10] Su, Z., Zhou, T., Li, Y., et al. Scaling Analysis of Liquid Lead-Bismuth Eutectic Thermal Stratification in ADS Facility Under LOCA. International Conference on Nuclear Engineering. (2013).

[11] Zhao, P.C, Zhu, E.P., Yu, H.X., et al. Research on Scaling Analysis Method for Natural Circulation Test Facility of Lead-Based Fast Reactor. Nuclear Power Engineering. (2020) 41(06), 207-213.

[12] Zhao, P.C, Wang, T., Zhu, E.P., et al. Research on proportional method for experimental facility of lead-based fast reactor with natural circulation. Ann. Nucl. Energy. (2020) 172, 109096.

[13] Li, Y., Lv, K.F., Chen, L.L., et al., Experiments and analysis on LBE steady natural circulation in a rectangular shape loop. Prog. Nucl. Energ. (2015) 81, 239-244.

[14] C. Fazio., V.P. Sobolev., A. Gessi., et al., Handbook on Lead-bismuth Eutectic Alloy and Lead Properties, Materials Compatibility, Thermal-hydraulics and Technologies: 2015 Edition. (2015).

[15] Chen, H.L., Chen, Z., Zhou, T., et al., Preliminary Thermal-hydraulic Design and Analysis of China Lead Alloy Cooled Research Reactor (CLEAR-I). Proceedings of the 9th International Topical Meeting on Nuclear Thermal-Hydraulics, Operation and Safety (NUTHOS-'12). 1-9 (2012).

[16] Yuan, L.F., Liu, Z.P., Wang, C.L., et al., Numerical study on thermal-hydraulic characteristics of lead-bismuth loop system under moving conditions. Ann. Nucl. Energy. (2024) 196: 110224.

[17] Cheng, C, Lu, D.H., Qianhua Su, Q.H., et al. Distortion Evaluation of Natural Circulation Characteristic Curves Based on Scaled-down Methodology. Nuclear Power Engineering. (2023) 44(06): 119-126.

[18] Wang, G.Q., Sun, D.B., Wang, J.C.,et al., Scaling analysis of single-phase natural circulation in deep-pool-type nuclear heating reactors. Prog. Nucl. Energ. (2025) 180, 105564.